

7_Concat&Merge

February 11, 2025

```
[1]: import pandas as pd
```

```
[2]: india_weather = pd.DataFrame({
      'city': ['mumbai', 'delhi', 'banglore'],
      'temperature': [32,45,30],
      'humidity': [80,60,78]
    })
india_weather
```

```
[2]:
```

	city	temperature	humidity
0	mumbai	32	80
1	delhi	45	60
2	banglore	30	78

```
[4]: us_weather = pd.DataFrame({
      'city': ['new york', 'chicago', 'orlando'],
      'temperature': [21,14,35],
      'humidity': [68,65,75]
    })
us_weather
```

```
[4]:
```

	city	temperature	humidity
0	new york	21	68
1	chicago	14	65
2	orlando	35	75

```
[5]: df = pd.concat([india_weather,us_weather])
```

```
[6]: df
```

```
[6]:
```

	city	temperature	humidity
0	mumbai	32	80
1	delhi	45	60
2	banglore	30	78
0	new york	21	68
1	chicago	14	65
2	orlando	35	75

```
[17]: df = pd.concat([india_weather,us_weather], ignore_index = True)
```

```
[18]: df
```

```
[18]:
```

	city	temperature	humidity
0	mumbai	32	80
1	delhi	45	60
2	banglore	30	78
3	new york	21	68
4	chicago	14	65
5	orlando	35	75

```
[20]: df = pd.concat([india_weather,us_weather], keys = ['india','us'])
```

```
[21]: df
```

```
[21]:
```

		city	temperature	humidity
india	0	mumbai	32	80
	1	delhi	45	60
	2	banglore	30	78
us	0	new york	21	68
	1	chicago	14	65
	2	orlando	35	75

```
[22]: df.loc['india']
```

```
[22]:
```

	city	temperature	humidity
0	mumbai	32	80
1	delhi	45	60
2	banglore	30	78

```
[25]: temperature_df = pd.DataFrame({
    'city':['mumbai', 'delhi', 'banglore'],
    'temperature': [32,45,30]
})
```

```
[28]: temperature_df = pd.DataFrame({
    'city': ['delhi', 'mumbai'],
    'windspeed': [7,12]
})
```

```
[29]: pd.concat([temperature_df,temperature_df],axis = 1)
```

```
[29]:
```

	city	temperature	city	temperature
0	mumbai	32	mumbai	32
1	delhi	45	delhi	45
2	banglore	30	banglore	30

```
[30]: df1 = pd.DataFrame({
      'city': ['new york', 'chicago', 'orlando'],
      'temperature': [21,14,35],
    })
```

```
[31]: df2 = pd.DataFrame({
      'city': ['new york', 'chicago', 'orlando'],
      'humidity': [68,65,75]
    })
```

```
[35]: df3 = pd.merge(df1,df2, on = 'city')
```

```
[36]: df3
```

```
[36]:
```

	city	temperature	humidity
0	new york	21	68
1	chicago	14	65
2	orlando	35	75

```
[45]: df1 = pd.DataFrame({
      'city': ['new york', 'chicago', 'orlando'],
      'temperature': [21,14,35],
    })
df1
```

```
[45]:
```

	city	temperature
0	new york	21
1	chicago	14
2	orlando	35

```
[46]: df2 = pd.DataFrame({
      'city': ['new york', 'chicago', 'virginia'],
      'humidity': [68,65,85]
    })
df2
```

```
[46]:
```

	city	humidity
0	new york	68
1	chicago	65
2	virginia	85

```
[47]: pd.merge(df1,df2, on = 'city',how = 'inner')
```

```
[47]:
```

	city	temperature	humidity
0	new york	21	68
1	chicago	14	65